

PROJECT KEYSTONE :– SERVICE MANAGEMENT PLATFORM

CERTIFICATE

This is to certify that "**Project KEYSTONE – Service Management Platform**" has been successfully completed by **Sujal Suresh Saptal** under the guidance of **ZIDIO Development** in partial fulfillment of the requirements for the Degree of Bachelor of Science in Information Technology.

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Chapter 1 – Introduction

KEYSTONE is a full-stack Field Service Management (FSM) platform developed to simplify maintenance operations for organizations. It provides an integrated solution for managing service requests, assigning technicians, tracking work orders, monitoring Service Level Agreements (SLAs), and improving communication between administrators, technicians, and customers. The system replaces traditional manual processes with a centralized digital platform, increasing operational efficiency, transparency, and customer satisfaction.

Chapter 2 – Problem Statement

Organizations face several challenges:

- Manual service request handling
- Delayed technician assignment
- Inventory mismanagement
- No real-time updates
- Difficulty in tracking service history
- Poor SLA monitoring
- Lack of centralized reporting

The proposed system eliminates these problems through automation.

Chapter 4 – Scope

The system can be used by:

- Service Companies
- Maintenance Departments
- IT Support Organizations
- Manufacturing Industries
- Customer Support Teams

The project supports:

- Managers
- Dispatchers
- Technicians
- Customers

Chapter 5 – Existing System

Existing systems are mostly manual.

Problems include:

- Paper records
- Slow communication
- Data duplication
- Inventory mismatch
- Human errors
- Difficult reporting

Chapter 6 – Proposed System

The proposed system automates the complete workflow.

Features include:

- Online work order creation
- Technician assignment
- Inventory tracking
- Customer portal
- JWT Authentication
- Dashboard analytics
- Live notifications
- SLA tracking
- File uploads
- Role-based access

Chapter 7 – Software Requirements

Backend

- Java 21
- Spring Boot 3.3.4
- Spring Security
- Spring Data JPA
- Flyway & Maven

Frontend

- React 18

- TypeScript
- Vite
- Axios
- React Router
- Recharts

Database

- MySQL 8

Other Tools

- Docker
- Docker Compose
- GitHub
- IntelliJ IDEA
- VS Code

Chapter 8 – Hardware Requirements

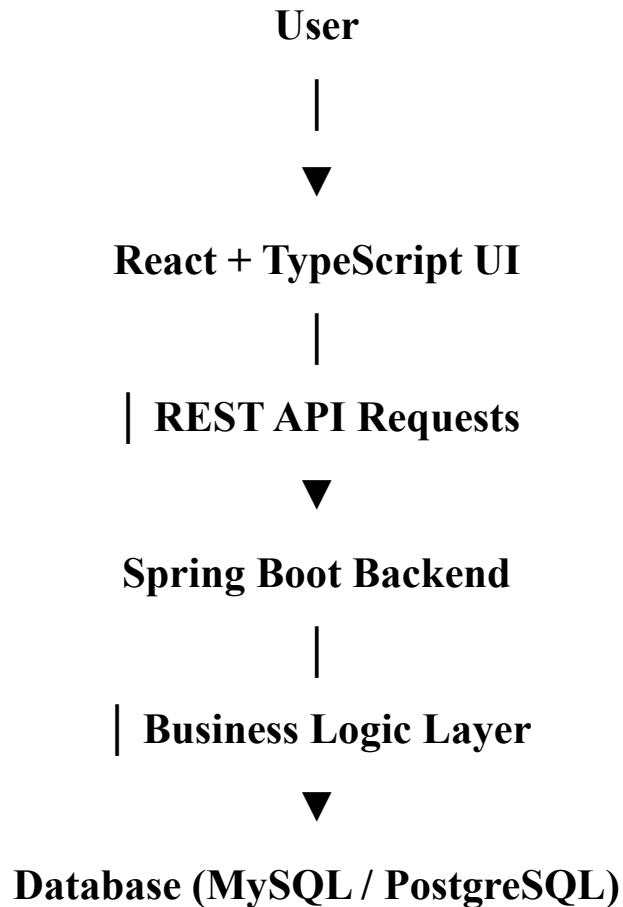
Minimum Requirements

- Intel i3 Processor
- 8 GB RAM
- 20 GB Storage
- Internet Connection

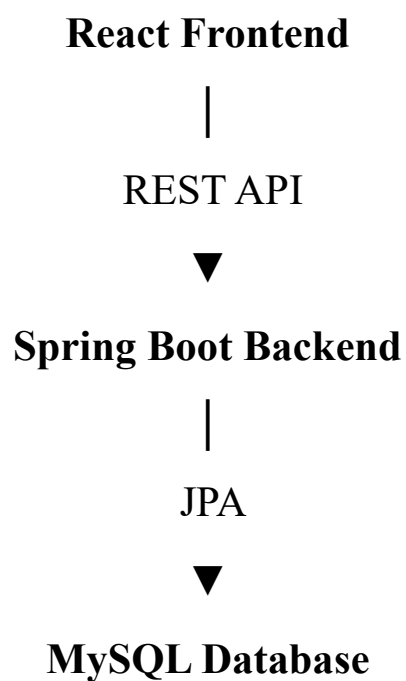
Recommended

- Intel i5/i7
- 16 GB RAM
- SSD Storage

Chapter 9 – System Architecture



Architecture follows a three-tier architecture.



Additional Components:

- JWT Authentication
- WebSocket
- Flyway Migration
- Docker Containers

Chapter 10 – Modules

1. Authentication Module

Functions:

- Login
- JWT Authentication
- Authorization
- User Roles

Roles:

- Manager
- Dispatcher
- Technician
- Customer

2. Customer Management

Functions:

- Add customer
- Update customer
- Search customer
- View history

3. Work Order Module

Functions:

- Create work order
- Assign technician
- Change status
- Attach files
- Close work order

Work Order States:-

NEW



ASSIGNED



IN_PROGRESS



ON_HOLD



COMPLETED



CLOSED

4. Technician Module

Functions:

- View assigned jobs
- Update status

- Upload images
- Record work time

5. Inventory Module

Functions:

- Manage parts
- Stock update
- Part usage
- Inventory reports

6. Dashboard

Displays:

- Open jobs
- Completed jobs
- SLA compliance
- Technician workload
- Priority distribution

7. Notification Module

Provides:

- Live notifications
- SLA alerts
- Assignment alerts
- Dashboard updates

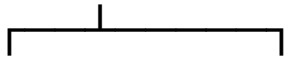
Chapter 11 – Database Design

Relationships

Customer



Work OrderR



Parts

Technician



Inventory

Database Relational Model & Entity Specifications

Entity	Primary Fields	Relationships & Constraints
User	id, email, password, name, role	Unique email; One-to-Many → WorkOrders, TimeLogs
Customer	id, name, contactEmail	One-to-Many → Sites
Site	id, name, address	Many-to-One → Customer; One-to-Many → WorkOrders
WorkOrder	id, code, title, status, priority	Many-to-One → Site, User; One-to-Many → TimeLogs, PartUsages
Part	id, name, sku, stockQty	Unique SKU; Inventory tracking
PartUsage	id, workOrderId, partId, quantity	Many-to-One → WorkOrder, Part
TimeLog	id, workOrderId, technicianId, minutesSpent	Many-to-One → WorkOrder, User

REST API Endpoint Specification

API Endpoint	Purpose
POST /api/auth/login	User login
POST /api/auth/register	Register user
GET /api/work-orders	View work orders
PUT /api/work-orders/{id}/status	Update status
PUT /api/work-orders/{id}/assign	Assign technician
POST /api/work-orders/{id}/time-log	Add time log
POST /api/work-orders/{id}/part-usage	Record parts used
GET /api/sites	View sites
GET / PUT /api/users/profile	Manage profile
GET /api/users/profile/history	View activity history

Chapter 12 – Technologies Used / Technology Stack

Project Name	KEYSTONE Field Service Platform
Domain	Enterprise Field Service Management (FSM) & SLA Compliance
Frontend Stack	React 18, TypeScript, Vite
Backend Stack	Java 21, Spring Boot 3.3(REST API)
Database	MySQL 8
Security	Spring Security with JWT Bearer Authentication
Version	1.0.0 – Production Release
Build Tool	Maven
Containerization	Docker & Docker Compose
API Type	RESTful APIs
Authentication	JWT (JSON Web Token)
Frontend Deployment	Vercel
Backend Deployment	Docker / Spring Boot Server
Live Web Link	https://keystone-service-management.vercel.app/login
GitHub Repository	https://github.com/ssaptal08-glitch/Keystone-ServiceManagement/tree/main/ServiceManagement

Chapter 13 – Implementation

Workflow:

Customer Login



Create Service Request



Dispatcher Reviews Request



Assign Technician



Technician Accepts Work



Inventory Used



Status Updated



Completed



Closed

Chapter 14 – Testing / Screenshots

Testing Performed:-

Unit Testing

- Spring Boot Tests
- JUnit
- Mockito

Integration Testing

- TestContainers
- MySQL

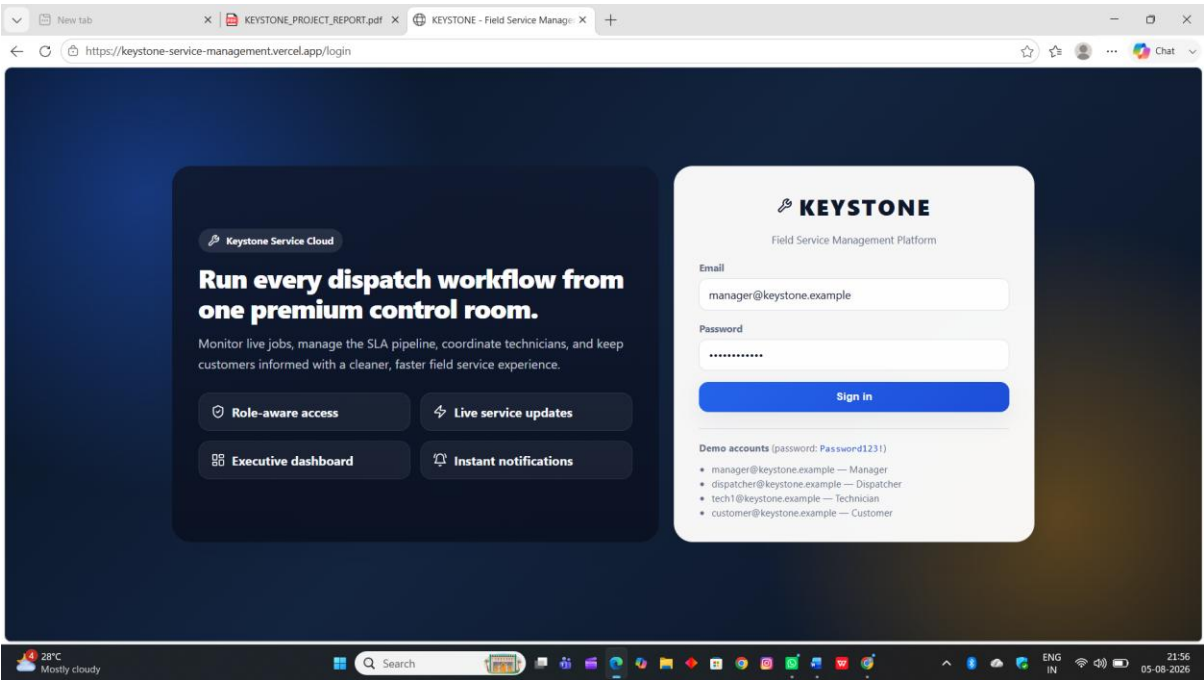
Frontend Testing

- API Testing
- UI Validation

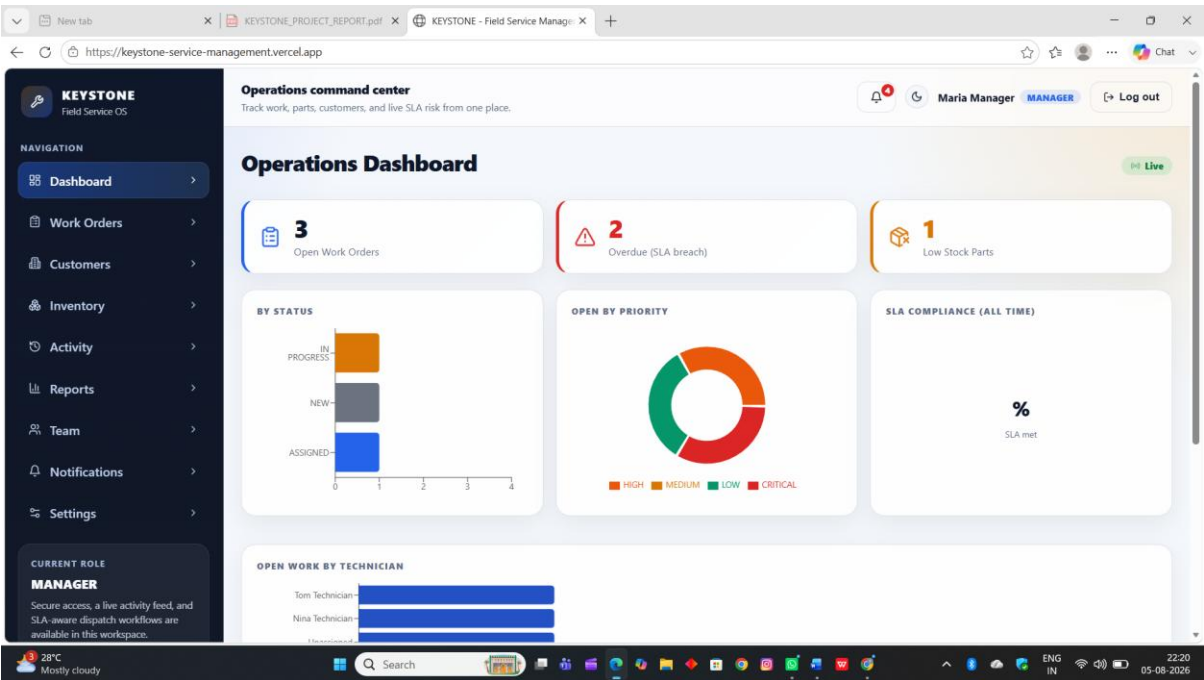
Manual Testing

- Login
- Dashboard
- CRUD Operations
- Notifications
- Inventory

Login



Dashboard



Work Order

The screenshot shows the 'Work Orders' page in the Keystone Field Service OS. The interface is in light mode. On the left is a dark sidebar with navigation links: Dashboard, Work Orders (selected), Customers, Inventory, Activity, Reports, Team, Notifications, and Settings. Below the sidebar, the user's role is listed as 'MANAGER'. The main content area has a header 'Operations command center' and a sub-header 'Work Orders'. It includes a search bar, a status dropdown set to 'All', and a table with columns: CODE, TITLE, CUSTOMER / SITE, STATUS, PRIORITY, TECHNICIAN, and DUE. The table is currently empty, displaying 'No work orders found.' A '+ New Work Order' button is in the top right. The bottom of the screen shows a Windows taskbar with a search bar and various application icons.

Inventory(DarK Mode feature)

The screenshot shows the 'Inventory' page in the Keystone Field Service OS, with the entire interface in dark mode. The sidebar on the left has the same navigation links as the previous screenshot, with 'Inventory' now selected. The main content area has a header 'Operations command center' and a sub-header 'Inventory'. It features a search bar and a 'Low stock only' toggle. Below this is a table with columns: SKU, NAME, UNIT COST, ON HAND, REORDER AT, and a 'Restock' button for each item. The table lists four items: MOT-4400 (1/2 HP Motor, \$145.00, 12 on hand, reorder at 3), BLT-2050 (Drive Belt 50in, \$22.00, 6 on hand, reorder at 5), FLT-3300 (HVAC Air Filter 20x20, \$14.75, 3 on hand, reorder at 5), and BRG-1001 (Sealed Ball Bearing 20mm, \$8.50, 40 on hand, reorder at 10). At the bottom of the main area is an 'ADD PART' form with fields for SKU, Name, Unit cost, and Initial qty. The bottom of the screen shows a Windows taskbar with a search bar and various application icons.

SKU	NAME	UNIT COST	ON HAND	REORDER AT	
MOT-4400	1/2 HP Motor	\$145.00	12	3	Restock
BLT-2050	Drive Belt 50in	\$22.00	6	5	Restock
FLT-3300	HVAC Air Filter 20x20	\$14.75	3	5	Restock
BRG-1001	Sealed Ball Bearing 20mm	\$8.50	40	10	Restock

Chapter 15 – Advantages

- Secure Login & Scalable Architecture
- Fast Processing
- Centralized Data
- Real-Time Updates
- Easy Tracking
- Better Customer Service
- Inventory Management
- Dashboard Analytics

Chapter 16 – Limitations

- Internet required
- Uses local storage for attachments
- MySQL only
- No mobile application
- Limited offline support

Chapter 17 – Future Enhancements

- Android Application
- iOS Application
- AI-based Technician Assignment
- GPS Tracking

- Email Notifications
- SMS Integration
- Cloud Storage
- Payment Gateway
- Predictive Maintenance
- Multi-language Support

Chapter 18 – Production Verification & Live Deployment Links

Target Environment	Live URL / Location
Vercel Live Frontend	https://keystone-service-management.vercel.app/login
GitHub Source Code	https://github.com/sssaptal08-glitch/Keystone-ServiceManagement/tree/main/ServiceManagement
Railway Backend	backend/Dockerfile, render.yaml https://keystone-servicemanagement-production.up.railway.app

Chapter 19 – Conclusion

The **KEYSTONE Field Service Platform** successfully automates the complete field service management process, including customer management, work order tracking, technician assignment, inventory management, and SLA monitoring. By using modern technologies such as React, Spring Boot, MySQL, and JWT authentication, the system provides a secure, efficient, and user-friendly solution. Overall, the project improves operational efficiency, reduces manual work, and offers a scalable platform for organizations to manage their field service operations effectively.

Chapter 20 – References

1. Oracle Java Documentation
2. Spring Boot Documentation
3. React Documentation
4. MySQL Documentation
5. Docker Documentation
6. Flyway Documentation
7. Maven Documentation
8. JWT Documentation
9. Vite Documentation
10. GitHub Documentation